Dr. Víctor Manuel Chávez Pérez

Professor–Researcher · Applied Physics, Data Science & Science Education

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Google Scholar · ResearchGate · LinkedIn

SNII · Candidate (SECIHTI) 2025–2029

PROFILE

I am a physicist from the University of Guadalajara and hold a PhD in Applied Physics from the University of Vigo. My research blends atmospheric and climate physics with large-scale data analysis: I study sudden stratospheric warmings (SSW), heat waves and their response under climate-change scenarios using models such as CMIP5 and WACCM.

EDUCATION

2017 – 2024	**PhD in Applied Physics · Universidade de Vigo**
2011 – 2012	**MSc in Climate Science: Meteorology, Physical Oceanography & Climate Change · Universidade de Vigo**
2009 – 2011	**MSc in Hydrometeorology — Meteorology & Physical Oceanography · Universidad de Guadalajara**
2000 – 2008	**BSc in Physics · Universidad de Guadalajara**

EXPERIENCE

2024 – present	**Lecturer "B" · Centro Universitario de Tlaquepaque · UdeG**
2025 – 2026	**Lecturer · Centro Universitario Bonpland y Humboldt**
2019 – 2026	**Adjunct Professor ("Profesor de Asignatura B") · Universidad Tecnológica de Jalisco**
2019 – 2020	**Faculty Lecturer · Tecmilenio · Campus Guadalajara**
2019	**Physics Instructor · UNITEC · Campus Guadalajara**

PUBLICATIONS

2025	Observación Astronómica en México: Pasando de lo Visible a lo Invisible — *Óptica Pura y Aplicada 58(1):15* · DOI
2025	Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations — *Atmosphere 16(5)* · DOI
2025	Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios — *Climate 13(10)* · DOI
2023	Estacionalidad y Anomalías del Nivel del Mar en el Caribe por Medio de Datos de Altimetría — *Generis Publishing*
2022	Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Sudden Stratospheric Warmings — *Atmosphere 13(4)* · DOI
2018	The climatology of dust events over the European continent using data of the BSC-DREAM8b model — *Atmospheric Research 209:144–162* · DOI

CONFERENCES & TALKS

2018	American Geophysical Union (AGU) Fall Meeting 2018 · Washington, USA
2017	European Geosciences Union (EGU) General Assembly 2017 · Vienna, Austria
2016	SPARC Workshop SHARP 2016 — Stratospheric Change and its Role for Climate Prediction · Berlin, Germany
2015	European Geosciences Union (EGU) General Assembly · Vienna, Austria
2015	SPARC Regional Workshop — Role of the stratosphere in climate variability and prediction · Granada, Spain

COMMUNITY BUILDING & SERVICE

2011 **Thesis supervision:** Estacionalidad y anomalías del nivel del mar en el Caribe por medio de datos de altimetría (Co-advisor)

Journal peer review: IEEE Latin America Transactions, Int. J. of Environment and Pollution (Inderscience), Congreso de Ingeniería, Ciencia y Gestión Ambiental (AMICA)
Project evaluation: External evaluator — 2026 National Call, Basic & Frontier Science (3 proposals)

RESEARCH INTERESTS

Sudden stratospheric warmings · Stratospheric dynamics · General circulation models (CMIP5 / WACCM) · Climate change · Physical oceanography & remote sensing · Applied optics · Data science · High-performance computing · Science education & outreach

LANGUAGES

Spanish — Native · English — Intermediate

TOOLBOX

Python · Fortran · MatLab · IDL · NCL · LaTeX · COMSOL · LabVIEW · MPI · Linux · Moodle · Pandas · Scikit-learn · Jupyter